



EXECUTIVE IPSEC SECURITY BRIEFING

# Executive IPsec VPN Security & Risk Posture

High-level strategic risk evaluation, cryptographic posture, and operational exposure briefing for leadership

## Section 1: Executive Summary & System Metadata

Session timestamps, endpoints evaluated, and overall audit verdict

### ATTENTION REQUIRED

YES · HIGH

Link 'Link 3: Field Recon <-> Mission Hub' demonstrates operational encryption via AES-128-CBC, but Perfect Forward Secrecy (PFS) is DISABLED on Child SAs, and TRANSPORT mode exposes subnet headers. Historic communications remain at risk if long-term credentials are ever breached.

#### SESSION TIMESTAMPS

**15.0 sec**

Live Link Telemetry: +0.0s to +15.0s (15.0 sec)

#### ORIGIN GATEWAY

**Link 3: Field Recon <-> Mission Hub  
(Initiator Node)**

192.168.10.12 • StrongSwan Client Daemon

#### SECURED TUNNEL

**IPsec TRANSPORT Tunnel • AES-128-CBC**

AES-128-CBC / Group 14 (MODP 2048)

#### REMOTE GATEWAY

**Enterprise Responder Node (Link 3: Field  
Recon <-> Mission Hub)**

10.200.0.1 • StrongSwan Server Gateway

## Section 2: Holistic Risk Scorecard

Visual 0–100 score gauge, NIST compliance status badge, and deployment readiness rating

#### COMPOSITE RISK INDEX



MODERATE RISK

#### NIST SP 800-77 REV. 1 COMPLIANCE

**CRITICAL  
FAIL**

NIST SP  
800-77  
Rev. 1

#### CRYPTOGRAPHIC RESILIENCE

**72%**

AES-128-CBC active, PFS  
missing

#### COMPLIANCE ALIGNMENT

**CONDITIONAL**

NIST SP 800-77 §4.2.3 Child SA  
gap

#### DEPLOYMENT READINESS RATING

**CONDITIONAL  
DEPLOYMENT**

Conditionally permitted on internal  
backbones; Child SA PFS enforcement  
and Tunnel Mode encapsulation required  
before external exposure.

#### THREAT EXPOSURE INDEX

**MODERATE**

Retroactive decryption &  
subnet leakage

#### CHANNEL STABILITY

**99.5%**

Monotonic sequence verification



THREAT EXPOSURE & BUSINESS IMPACT EVALUATION

# Executive Threat Matrix

Comprehensive register mapping discovered cryptographic and protocol vulnerabilities to direct business risk

Section 3: Executive Threat Matrix

Grid mapping critical vulnerabilities to business risk and operational exposure vectors

|                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><h3>Retrospective Bulk Decryption</h3><div>Sev: HIGHLikelihood: Moderate</div><p>High risk. Ephemeral DH exchange omitted during Child SA rekeying; if long-term credentials leak, all historic traffic can be decrypted.</p><div>ACTIVE GAP</div></div> | <div><h3>Subnet Architecture &amp; Metadata Exposure</h3><div>Sev: MEDIUMLikelihood: High</div><p>Moderate risk. Internal IP headers exposed on transit network, revealing internal network topologies to wiretappers.</p><div>ACTIVE GAP</div></div> |
| <div><h3>Padding Oracle Interception</h3><div>Sev: MEDIUMLikelihood: Moderate</div><p>Moderate risk. CBC ciphers rely on separate HMAC and are susceptible to padding oracle attacks if not strictly sequenced.</p><div>OBSERVED</div></div>                  | <div><h3>Replay Injection &amp; Sequence Tampering</h3><div>Sev: LOWLikelihood: Low</div><p>Zero risk. RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted across connection lifetime.</p><div>PROTECTED</div></div>                     |



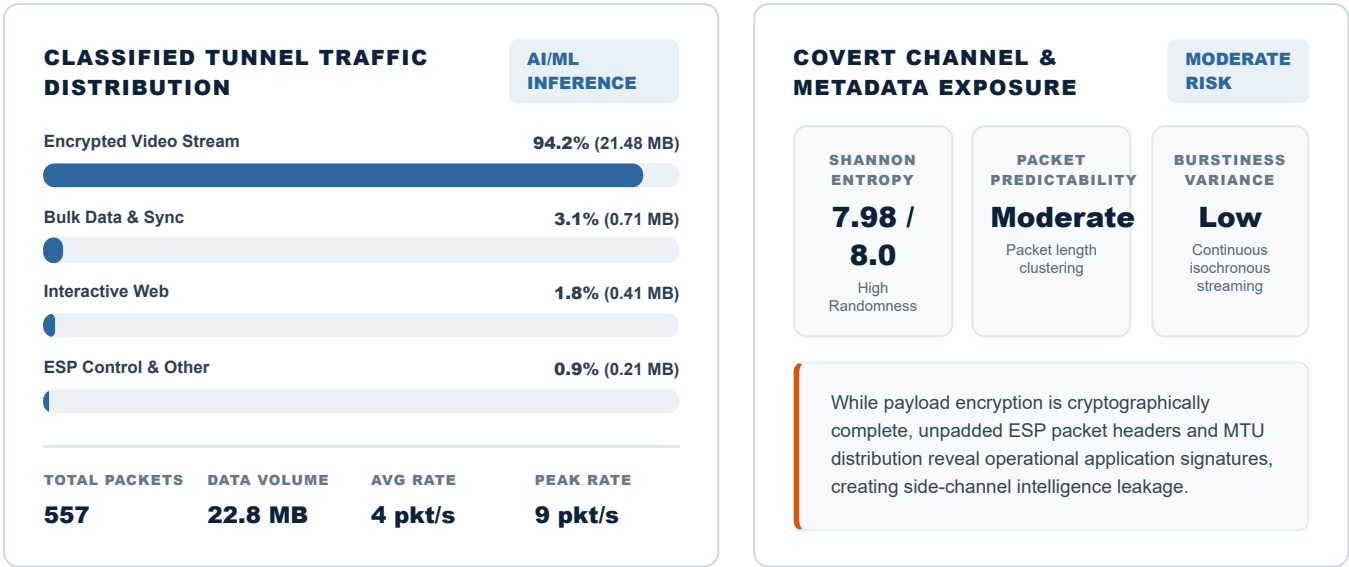
TRAFFIC CLASSIFICATION & STRATEGIC ROADMAP

## Traffic Overview & Strategic Remediation

Deep payload inference distribution, side-channel leakage, and prioritized resource allocation roadmap

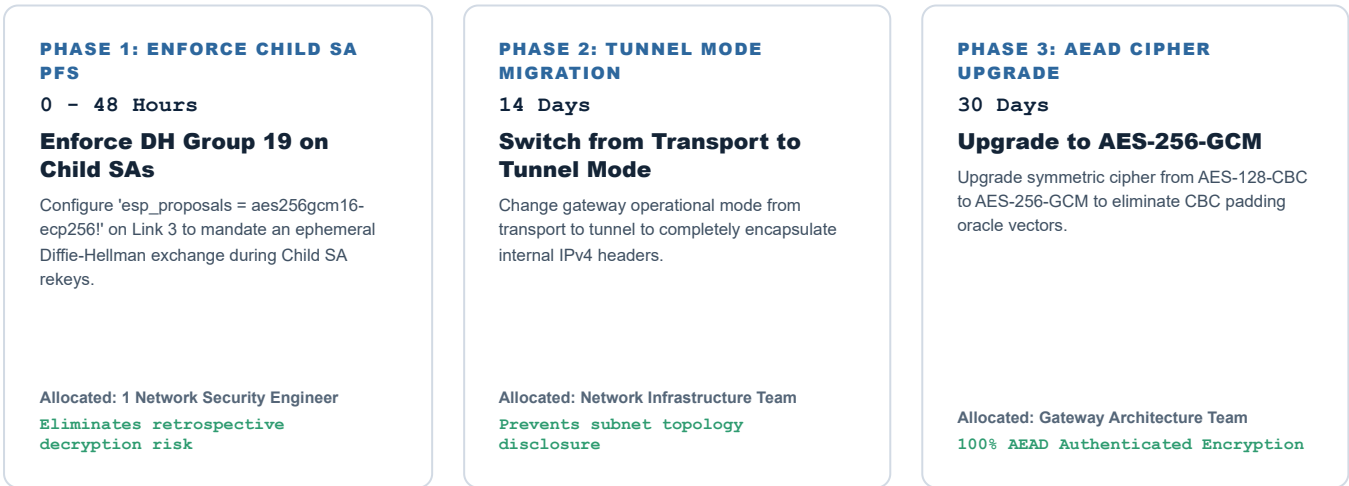
### Section 4: High-Level Traffic Overview

Summary distribution of classified tunnel traffic and covert channel risk



### Section 5: Strategic Remediation Action Plan

Prioritized recommendations for leadership sign-off and operational resource allocation



### Executive Sign-Off & Platform Attestation



